UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
COURSE CODE	:	**CSC429**
EXAMINATION	:	**FEBRUARY 2023**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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PART A

QUESTION 1

List the name of the inventor of the following devices:
a) Mechanical Calculating Machine
b) Difference Machine
c) Analytical Machine
d) IAS Machine
e) Integrated circuit

(5 marks)

QUESTION 2

a) Identify the Von Neumann architecture input and output components based on the excerpts from the web.

> The Jacquard Loom is a mechanical loom invented by Joseph Marie Jacquard in 1801, which simplifies the process of manufacturing textiles with complex patterns. The loom is controlled by punched cards with punched holes, each row corresponding to one design row. Multiple rows of holes are punched on each card, and the many cards that compose the textile design are strung together in an orderly manner.

Figure 1: Excerpt from https://en.wikipedia.org/wiki/Loom

(2 marks)

b) Identify **ONE (1)** similarity and **ONE (1)** dissimilarity between the mechanical calculating machine and Atanasoff Berry Computer.

(3 marks)

QUESTION 3

Solve the following equations in binary number bases. Show your workflow.

a) $1.0000\ 1001_2 - 0.0011\ 0011_2$
b) $0.FA3_{16} - 0.651_8$

(5 marks)

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QUESTION 4

What is the decimal value of the sum of the binary numbers 1101 1100 and 0010 0111 if they are represented in sign and magnitude? Assume that the basic unit is the byte. Show your workflow.

(5 marks)

QUESTION 5

a) Convert 43.046875_{10} to a floating point number in IEEE single precision format in hexadecimal value form.

(5 marks)

b) Convert C478 0000h in IEEE single precision floating point number to decimal value form.

(5 marks)

QUESTION 6

Describe the function of each register in the computer system.

(10 marks)

QUESTION 7

a) Differentiate SRAM and DRAM from the perspective of the following characteristics.
   i) The way the bits are stored.
   ii) Speed.
   iii) Location it is being used.

(6 marks)

b) Why does DRAM need refreshing?

(2 marks)

c) Comparing DRAM and SRAM, which memory would be placed higher in the memory hierarchy? Give your justification to support your answer.

(2 marks)

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QUESTION 8

Question 8 is based on Figure 2.

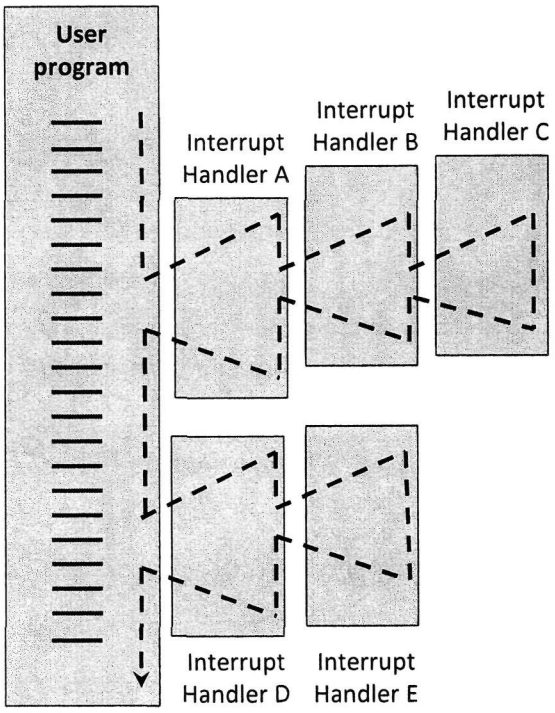


Figure 2: Multiple interrupts diagram

a) Describe maskable interrupt.

(2 marks)

b) What is the sequence of the program execution if the defined priority is implemented in Figure 2?

(4 marks)

c) What is the sequence of the program execution if the disable interrupt is implemented in Figure 2?

(2 marks)

d) Which of the interruption method between defined priority or disable interrupt can finish Interrupt C earlier? Give your justification.

(2 marks)

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QUESTION 9

a) Discuss the **THREE (3)** input-output techniques available in the computer system.
(6 marks)

b) Universal Serial Bus (USB) has a "hot connection" to the devices. Briefly explain what is meant by "hot connection."
(2 marks)

c) Why is there a need for the I/O module to be the mediator between the CPU and devices?
(2 marks)

QUESTION 10

a) Given the architecture of Single Instruction, Single Data Stream (SISD) is in Figure 3, with CU representing Control Unit, PU representing the Processing Unit, and MU representing the Memory Unit. Draw the following multiple processor organization with an appropriate label. Clearly show the Instruction Stream (IS) and Data Stream (DS).
i) Multiple Instructions Multiple Data streams (MIMD) with shared memory.
ii) Multiple Instructions Multiple Data streams (MIMD) with distributed memory.
(6 marks)

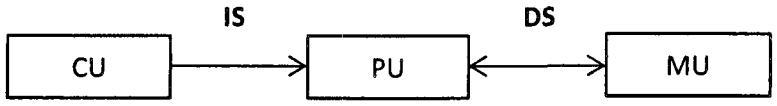


Figure 3: Single Instruction, Single Data Stream (SISD)

b) Give **ONE (1)** example for the following multiple-processor organization.
i) Multiple Instructions Multiple Data streams (MIMD) with shared memory.
ii) Multiple Instructions Multiple Data streams (MIMD) with distributed memory.
(2 marks)

c) Discuss **ONE (1)** advantage of using Multiple Instructions Multiple Data stream (MIMD) with distributed memory.
(2 marks)

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PART B

QUESTION 1

Question 1 is based on Figure 4 and Table 3.

$$z = A + B^2 - C\,(D - C / B)$$

Figure 4

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 4 using zero-, one-, and two-address instructions. The instructions available for use are provided in Table 1. Note that Register ***A***, ***B, C,*** and ***D*** cannot be modified, and Register ***x, y***, and ***z*** can be used as temporary registers. However, the final answer must be stored in Register ***z***.

(20 marks)

Table 3: Machine Instruction

Zero-address	One-address	Two-address
PUSH x	LOAD x	MOVE (x,y)
POP x	STOR x	ADD (x,y)
ADD	ADD x	SUB (x,y)
SUB	SUB x	MUL (x,y)
MUL	MUL x	DIV (x,y)
DIV	DIV x	

END OF QUESTION PAPER

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